
ENGINEERING MATHEMATICS

ANALYTIC FUNCTIONS

UNIT- III

ANALYTIC FUNCTIONS

3.1 INTRODUCTION

The theory of functions of a complex variable is the most important in solving a large number of Engineering and Science problems. Many complicated integrals of real function are solved with the help of a complex variable.

3.1 (a) Complex Variable

$x + iy$ is a complex variable and it is denoted by z .

(i. e.) $z = x + iy$ where $i = \sqrt{-1}$

3.1 (b) Function of a complex Variable

If $z = x + iy$ and $w = u + iv$ are two complex variables, and if for each value of z in a given region R of complex plane there corresponds one or more values of w is said to be a function z and is denoted by $w = f(z) = f(x + iy) = u(x, y) + iv(x, y)$ where $u(x, y)$ and $v(x, y)$ are real functions of the real variables x and y .

Note:

(i) single-valued function

If for each value of z in R there is correspondingly only one value of w , then w is called a single valued function of z .

Example: $w = z^2, w = \frac{1}{z}$

$w = z^2$					$w = \frac{1}{z}$				
z	1	2	-2	3	z	1	2	-2	3
w	1	4	4	9	w	1	$\frac{1}{2}$	$\frac{1}{-2}$	$\frac{1}{3}$

(ii) Multiple – valued function

If there is more than one value of w corresponding to a given value of z then w is called multiple – valued function.

Example: $w = z^{1/2}$

$w = z^{1/2}$			
z	4	9	1
w	-2, 2	3, -3	1, -1

(iii) The distance between two points z and z_o is $|z - z_o|$

(iv) The circle C of radius δ with centre at the point z_o can be represented by $|z - z_o| = \delta$.

(v) $|z - z_o| < \delta$ represents the interior of the circle excluding its circumference.

(vi) $|z - z_o| \leq \delta$ represents the interior of the circle including its circumference.

(vii) $|z - z_0| > \delta$ represents the exterior of the circle.

(viii) A circle of radius 1 with centre at origin can be represented by $|z| = 1$

3.1 (c) Neighbourhood of a point z_0

Neighbourhood of a point z_0 , we mean a sufficiently small circular region [excluding the points on the boundary] with centre at z_0 .

$$(i. e.) |z - z_0| < \delta$$

Here, δ is an arbitrary small positive number.

3.1 (d) Limit of a Function

Let $f(z)$ be a single valued function defined at all points in some neighbourhood of point z_0 .

Then the limit of $f(z)$ as z approaches z_0 is w_0 .

$$(i. e.) \lim_{z \rightarrow z_0} f(z) = w_0$$

3.1 (e) Continuity

If $f(z)$ is said to be continuous at $z = z_0$ then

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

If two functions are continuous at a point their sum, difference and product are also continuous at that point, their quotient is also continuous at any such point [$dr \neq 0$]

Example: 3.1 State the basic difference between the limit of a function of a real variable and that of a complex variable.

Solution:

In real variable, $x \rightarrow x_0$ implies that x approaches x_0 along the X-axis (or) a line parallel to the X-axis.

In complex variables, $z \rightarrow z_0$ implies that z approaches z_0 along any path joining the points z and z_0 that lie in the z -plane.

3.1 (f) Differentiability at a point

A function $f(z)$ is said to be differentiable at a point, $z = z_0$ if the limit

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \text{ exists.}$$

This limit is called the derivative of $f(z)$ at the point $z = z_0$

If $f(z)$ is differentiable at z_0 , then $f(z)$ is continuous at z_0 . This is the necessary condition for differentiability.

Example: 3.2 If $f(z)$ is differentiable at z_0 , then show that it is continuous at that point.

Solution:

As $f(z)$ is differentiable at z_0 , both $f(z_0)$ and $f'(z_0)$ exist finitely.

$$\text{Now, } \lim_{z \rightarrow z_0} |f(z) - f(z_0)| = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} (z - z_0)$$

$$\begin{aligned}
&= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \lim_{z \rightarrow z_0} (z - z_0) \\
&= f'(z_0) \cdot 0 = 0
\end{aligned}$$

$$\text{Hence, } \lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} f(z_0) = f(z_0)$$

As $f(z_0)$ is a constant.

This is exactly the statement of continuity of $f(z)$ at z_0 .

Example: 3.3 Give an example to show that continuity of a function at a point does not imply the existence of derivative at that point.

Solution:

$$\text{Consider the function } w = |z|^2 = x^2 + y^2$$

This function is continuous at every point in the plane, being a continuous function of two real variables. However, this is not differentiable at any point other than origin.

Example: 3.4 Show that the function $f(z)$ is discontinuous at $z = 0$, given that $f(z) = \frac{2xy^2}{x^2 + 3y^4}$, when $z \neq 0$ and $f(0) = 0$.

Solution:

$$\text{Given } f(z) = \frac{2xy^2}{x^2 + 3y^4},$$

$$\text{Consider } \lim_{z \rightarrow z_0} [f(z)] = \lim_{\substack{y=mx \\ x \rightarrow 0}} [f(z)] = \lim_{x \rightarrow 0} \frac{2x(mx)^2}{x^2 + 3(mx)^4} = \lim_{x \rightarrow 0} \left[\frac{2m^2x}{1 + 3m^4x^2} \right] = 0$$

$$\lim_{\substack{y^2=x \\ x \rightarrow 0}} [f(z)] = \lim_{x \rightarrow 0} \frac{2x^2}{x^2 + 3x^2} = \lim_{x \rightarrow 0} \frac{2x^2}{4x^2} = \frac{2}{4} = \frac{1}{2} \neq 0$$

$\therefore f(z)$ is discontinuous

Example: 3.5 Show that the function $f(z)$ is discontinuous at the origin ($z = 0$), given that

$$f(z) = \frac{xy(x-2y)}{x^3+y^3}, \text{ when } z \neq 0$$

$$= 0, \text{ when } z = 0$$

Solution:

$$\begin{aligned}
\text{Consider } \lim_{z \rightarrow z_0} [f(z)] &= \lim_{\substack{y=mx \\ x \rightarrow 0}} [f(z)] = \lim_{x \rightarrow 0} \frac{x(mx)(x-2(mx))}{x^3+(mx)^3} \\
&= \lim_{x \rightarrow 0} \frac{m(1-2m)x^3}{(1+m^3)x^3} = \frac{m(1-2m)}{1+m^3}
\end{aligned}$$

Thus $\lim_{z \rightarrow 0} f(z)$ depends on the value of m and hence does not take a unique value.

$\therefore \lim_{z \rightarrow 0} f(z)$ does not exist.

$\therefore f(z)$ is discontinuous at the origin.

3.2 ANALYTIC FUNCTIONS – NECESSARY AND SUFFICIENT CONDITIONS FOR ANALYTICITY IN CARTESIAN AND POLAR CO-ORDINATES

Analytic [or] Holomorphic [or] Regular function

A function is said to be analytic at a point if its derivative exists not only at that point but also in some neighbourhood of that point.

Entire Function: [Integral function]

A function which is analytic everywhere in the finite plane is called an entire function.

An entire function is analytic everywhere except at $z = \infty$.

Example: $e^z, \sin z, \cos z, \sinh z, \cosh z$

3.2 (i) The necessary condition for $f = (z)$ to be analytic. [Cauchy – Riemann Equations]

The necessary conditions for a complex function $f = (z) = u(x, y) + iv(x, y)$ to be analytic in a region R are $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ i.e., $u_x = v_y$ and $v_x = -u_y$

[OR]

Derive C – R equations as necessary conditions for a function $w = f(z)$ to be analytic.

Proof:

Let $f(z) = u(x, y) + iv(x, y)$ be an analytic function at the point z in a region R. Since $f(z)$ is analytic, its derivative $f'(z)$ exists in R

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$\text{Let } z = x + iy$$

$$\Rightarrow \Delta z = \Delta x + i\Delta y$$

$$z + \Delta z = (x + \Delta x) + i(y + \Delta y)$$

$$f(z) = u(x, y) + iv(x, y)$$

$$f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)$$

$$f(z + \Delta z) - f(z) = u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - [u(x, y) + iv(x, y)]$$

$$= [u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)]$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{u(x+\Delta x, y+\Delta y) - u(x, y) + i[v(x+\Delta x, y+\Delta y) - v(x, y)]}{\Delta x + i\Delta y}$$

Case (i)

If $\Delta z \rightarrow 0$, first we assume that $\Delta y = 0$ and $\Delta x \rightarrow 0$.

$$\begin{aligned} \therefore f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{[u(x+\Delta x, y) - u(x, y)] + i[v(x+\Delta x, y) - v(x, y)]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x+\Delta x, y) - v(x, y)}{\Delta x} \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad \dots (1) \end{aligned}$$

Case (ii)

If $\Delta z \rightarrow 0$ Now, we assume that $\Delta x = 0$ and $\Delta y \rightarrow 0$

$$\begin{aligned}\therefore f'(z) &= \lim_{\Delta y \rightarrow 0} \frac{[u(x, y+\Delta y) - u(x, y)] + i[v(x, y+\Delta y) - v(x, y)]}{i\Delta y} \\ &= \frac{1}{i} \lim_{\Delta y \rightarrow 0} \frac{u(x, y+\Delta y) - u(x, y)}{\Delta y} + \lim_{\Delta y \rightarrow 0} \frac{v(x, y+\Delta y) - v(x, y)}{\Delta y} \\ &= \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \\ &= -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \quad \dots (2)\end{aligned}$$

From (1) and (2), we get

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

Equating the real and imaginary parts we get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$(i.e.) u_x = v_y, \quad v_x = -u_y$$

The above equations are known as Cauchy – Riemann equations or C-R equations.

Note: (i) The above conditions are not sufficient for $f(z)$ to be analytic. The sufficient conditions are given in the next theorem.

(ii) Sufficient conditions for $f(z)$ to be analytic.

If the partial derivatives u_x, u_y, v_x and v_y are all continuous in D and $u_x = v_y$ and $u_y = -v_x$, then the function $f(z)$ is analytic in a domain D.

(ii) Polar form of C-R equations

In Cartesian co-ordinates any point z is $z = x + iy$.

In polar co-ordinates, $z = re^{i\theta}$ where r is the modulus and θ is the argument.

Theorem: If $f(z) = u(r, \theta) + iv(r, \theta)$ is differentiable at $z = re^{i\theta}$, then $u_r = \frac{1}{r}v_\theta$, $v_r = -\frac{1}{r}u_\theta$

$$(OR) \quad \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

Proof:

Let $z = re^{i\theta}$ and $w = f(z) = u + iv$

$$(i.e.) u + iv = f(z) = f(re^{i\theta})$$

Diff. p.w. r. to r , we get

$$\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} = f'(re^{i\theta}) e^{i\theta} \quad \dots (1)$$

Diff. p.w. r. to θ , we get

$$\frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} = f'(re^{i\theta}) e^{i\theta} \quad \dots (2)$$

$$= ri[f'(re^{i\theta}) e^{i\theta}]$$

$$= ri \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right] \text{ by (1)}$$

$$= ri \frac{\partial u}{\partial r} - r \frac{\partial v}{\partial r}$$

Equating the real and imaginary parts, we get

$$\frac{\partial u}{\partial \theta} = -i \frac{\partial v}{\partial r}, \quad \frac{\partial v}{\partial \theta} = r \frac{\partial u}{\partial r}$$

$$(i.e.) \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = \frac{-1}{r} \frac{\partial u}{\partial \theta}$$

Problems based on Analytic functions – necessary conditions Cauchy – Riemann equations

Example: 3.6 Show that the function $f(z) = xy + iy$ is continuous everywhere but not differentiable anywhere.

Solution:

Given $f(z) = xy + iy$

(i.e.) $u = xy, v = y$

x and y are continuous everywhere and consequently $u(x, y) = xy$ and $v(x, y) = y$ are continuous everywhere.

Thus $f(z)$ is continuous everywhere.

But

$u = xy$	$v = y$
$u_x = y$	$v_x = 0$
$u_y = x$	$v_y = 1$
$u_x \neq v_y$	$u_y \neq -v_x$

C–R equations are not satisfied.

Hence, $f(z)$ is not differentiable anywhere though it is continuous everywhere .

Example: 3.7 Show that the function $f(z) = \bar{z}$ is nowhere differentiable.

Given $f(z) = \bar{z} = x - iy$

Solution:

$u = x$	$v = -y$
$\frac{\partial u}{\partial x} = 1$	$\frac{\partial v}{\partial x} = 0$
$\frac{\partial u}{\partial y} = 0$	$\frac{\partial v}{\partial y} = -1$

$$\therefore u_x \neq v_y$$

C–R equations are not satisfied anywhere.

Hence, $f(z) = \bar{z}$ is not differentiable anywhere (or) nowhere differentiable.

Example: 3.8 Show that $f(z) = |z|^2$ is differentiable at $z = 0$ but not analytic at $z = 0$.

Solution:

Let $z = x + iy$

$$\bar{z} = x - iy$$

$$|z|^2 = z \bar{z} = x^2 + y^2$$

$$(i.e.) f(z) = |z|^2 = (x^2 + y^2) + i0$$

$u = x^2 + y^2$	$v = 0$
$u_x = 2x$	$v_x = 0$
$u_y = 2y$	$v_y = 0$

So, the C-R equations $u_x = v_y$ and $u_y = -v_x$ are not satisfied everywhere except at $z = 0$.

So, $f(z)$ may be differentiable only at $z = 0$.

Now, $u_x = 2x$, $u_y = 2y$, $v_x = 0$ and $v_y = 0$ are continuous everywhere and in particular at $(0,0)$.

Hence, the sufficient conditions for differentiability are satisfied by $f(z)$ at $z = 0$.

So, $f(z)$ is differentiable at $z = 0$ only and is not analytic there.

Inverse function

Let $w = f(z)$ be a function of z and $z = F(w)$ be its inverse function.

Then the function $w = f(z)$ will cease to be analytic at $\frac{dz}{dw} = 0$ and $z = F(w)$ will be so, at point where

$$\frac{dw}{dz} = 0.$$

Example: 3.9 Show that $f(z) = \log z$ analytic everywhere except at the origin and find its derivatives.

Solution:

$$\text{Let } z = re^{i\theta}$$

$$f(z) = \log z$$

$$= \log(re^{i\theta}) = \log r + \log(e^{i\theta}) = \log r + i\theta$$

But, at the origin, $r = 0$. Thus, at the origin,

$$f(z) = \log 0 + i\theta = -\infty + i\theta$$

$$\text{Note : } e^{-\infty} = 0$$

$$\log e^{-\infty} = \log 0 ; -\infty = \log 0$$

So, $f(z)$ is not defined at the origin and hence is not differentiable there.

At points other than the origin, we have

$u(r, \theta) = \log r$	$v(r, \theta) = \theta$
$u_r = \frac{1}{r}$	$v_r = 0$
$u_\theta = 0$	$v_\theta = 1$

So, $\log z$ satisfies the C-R equations.

Further $\frac{1}{r}$ is not continuous at $z = 0$.

So, $u_r, u_\theta, v_r, v_\theta$ are continuous everywhere except at $z = 0$. Thus $\log z$ satisfies all the sufficient conditions for the existence of the derivative except at the origin. The derivative is

$$f'(z) = \frac{u_r + iv_r}{e^{i\theta}} = \frac{\left(\frac{1}{r}\right) + i(0)}{e^{i\theta}} = \frac{1}{re^{i\theta}} = \frac{1}{z}$$

Note: $f(z) = u + iv \Rightarrow f(re^{i\theta}) = u + iv$

Differentiate w.r.to 'r', we get

$$(i.e.) e^{i\theta} f'(re^{i\theta}) = \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r}$$

Example: 3.10 Check whether $w = \bar{z}$ is analytics everywhere.

Solution:

$$\text{Let } w = f(z) = \bar{z}$$

$$u+iv = x - iy$$

$u = x$	$v = -y$
$u_x = 1$	$v_x = 0$
$u_y = 0$	$v_y = -1$

$$u_x \neq v_y \text{ at any point } p(x,y)$$

Hence, C–R equations are not satisfied.

$\therefore$ The function $f(z)$ is nowhere analytic.

Example: 3.11 Test the analyticity of the function $w = \sin z$.

Solution:

$$\text{Let } w = f(z) = \sin z$$

$$u + iv = \sin(x + iy)$$

$$u + iv = \sin x \cos iy + \cos x \sin iy$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

Equating real and imaginary parts, we get

$u = \sin x \cosh y$	$v = \cos x \sinh y$
$u_x = \cos x \cosh y$	$v_x = -\sin x \sinh y$
$u_y = \sin x \sinh y$	$v_y = \cos x \cosh y$

$$\therefore u_x = v_y \text{ and } u_y = -v_x$$

C –R equations are satisfied.

Also the four partial derivatives are continuous.

Hence, the function is analytic.

Example: 3.12 Determine whether the function $2xy + i(x^2 - y^2)$ is analytic or not.

Solution:

$$\text{Let } f(z) = 2xy + i(x^2 - y^2)$$

(i.e.)	$u = 2xy$	$v = x^2 - y^2$
	$\frac{\partial u}{\partial x} = 2y$	$\frac{\partial v}{\partial x} = 2x$
	$\frac{\partial u}{\partial y} = 2x$	$\frac{\partial v}{\partial y} = -2y$

$$u_x \neq v_y \text{ and } u_y \neq -v_x$$

C–R equations are not satisfied.

Hence, $f(z)$ is not an analytic function.

Example: 3.13 Prove that $f(z) = \cosh z$ is an analytic function and find its derivative.

Solution:

$$\begin{aligned}\text{Given } f(z) &= \cosh z = \cos(iz) = \cos[i(x + iy)] \\ &= \cos(ix - y) = \cos ix \cos y + \sin(ix) \sin y \\ u + iv &= \cosh x \cos y + i \sinh x \sin y\end{aligned}$$

$u = \cosh x \cos y$	$v = \sinh x \sin y$
$u_x = \sinh x \cos y$	$v_x = \cosh x \sin y$
$u_y = -\cosh x \sin y$	$v_y = \sinh x \cos y$

$\therefore u_x, u_y, v_x$ and v_y exist and are continuous.

$$u_x = v_y \text{ and } u_y = -v_x$$

C–R equations are satisfied.

$\therefore f(z)$ is analytic everywhere.

$$\begin{aligned}\text{Now, } f'(z) &= u_x + iv_x \\ &= \sinh x \cos y + i \cosh x \sin y \\ &= \sinh(x + iy) = \sinh z\end{aligned}$$

Example: 3.14 If $w = f(z)$ is analytic, prove that $\frac{dw}{dz} = \frac{\partial w}{\partial x} = -i \frac{\partial w}{\partial y}$ where $z = x + iy$, and prove that

$$\frac{\partial^2 w}{\partial z \partial \bar{z}} = 0. \quad [\text{Anna, Nov 2001}]$$

Solution:

$$\text{Let } w = u(x, y) + iv(x, y)$$

As $f(z)$ is analytic, we have $u_x = v_y, u_y = -v_x$

$$\begin{aligned}\text{Now, } \frac{dw}{dz} &= f'(z) = u_x + iv_x = v_y - iu_y = i(u_y + iv_y) \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -i \left[\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right] \\ &= \frac{\partial}{\partial x} (u + iv) = -i \frac{\partial}{\partial y} (u + iv) \\ &= \frac{\partial w}{\partial x} = -i \frac{\partial w}{\partial y}\end{aligned}$$

We know that, $\frac{\partial w}{\partial z} = 0$

$$\therefore \frac{\partial^2 w}{\partial z \partial \bar{z}} = 0$$

$$\text{Also } \frac{\partial^2 w}{\partial \bar{z} \partial z} = 0$$

Example: 3.15 Prove that every analytic function $w = u(x, y) + iv(x, y)$ can be expressed as a function of z alone.

Proof:

$$\text{Let } z = x + iy \quad \text{and} \quad \bar{z} = x - iy$$

$$x = \frac{z + \bar{z}}{2} \quad \text{and} \quad y = \frac{z - \bar{z}}{2i}$$

Hence, u and v and also w may be considered as a function of z and $\bar{z}$

$$\begin{aligned} \text{Consider } \frac{\partial w}{\partial \bar{z}} &= \frac{\partial u}{\partial \bar{z}} + i \frac{\partial v}{\partial \bar{z}} \\ &= \left(\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \bar{z}} \right) + \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial \bar{z}} \right) \\ &= \left(\frac{1}{2} u_x - \frac{1}{2i} u_y \right) + i \left(\frac{1}{2} v_x - \frac{1}{2i} v_y \right) \\ &= \frac{1}{2} (u_x - v_y) + \frac{i}{2} (u_y + v_x) \\ &= 0 \text{ by C-R equations as } w \text{ is analytic.} \end{aligned}$$

This means that w is independent of $\bar{z}$

(i.e.) w is a function of z alone.

This means that if $w = u(x, y) + iv(x, y)$ is analytic, it can be rewritten as a function of $(x + iy)$.

Equivalently a function of $\bar{z}$ cannot be an analytic function of z .

Example: 3.16 Find the constants a, b, c if $f(z) = (x + ay) + i(bx + cy)$ is analytic.

Solution:

$$\begin{aligned} f(z) &= u(x, y) + iv(x, y) \\ &= (x + ay) + i(bx + cy) \end{aligned}$$

$u = x + ay$	$v = bx + cy$
$u_x = 1$	$v_x = b$
$u_y = a$	$v_y = c$

Given $f(z)$ is analytic

$$\Rightarrow u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$1 = c \quad \text{and} \quad a = -b$$

Example: 3.17 Examine whether the following function is analytic or not $f(z) = e^{-x}(\cos y - i \sin y)$.

Solution:

$$\begin{aligned} \text{Given } f(z) &= e^{-x}(\cos y - i \sin y) \\ \Rightarrow u + iv &= e^{-x} \cos y - ie^{-x} \sin y \end{aligned}$$

$u = e^{-x} \cos y$	$v = -e^{-x} \sin y$
$u_x = -e^{-x} \cos y$	$v_x = e^{-x} \sin y$
$u_y = -e^{-x} \sin y$	$v_y = -e^{-x} \cos y$

Here, $u_x = v_y$ and $u_y = -v_x$

$\Rightarrow$ C-R equations are satisfied

$\Rightarrow f(z)$ is analytic.

Example: 3.18 Test whether the function $f(z) = \frac{1}{2} \log(x^2 + y^2 + i \tan^{-1}(\frac{y}{x}))$ is analytic or not.

Solution:

Given $f(z) = \frac{1}{2} \log(x^2 + y^2 + i \tan^{-1}(\frac{y}{x}))$

(i.e.) $u + iv = \frac{1}{2} \log(x^2 + y^2 + i \tan^{-1}(\frac{y}{x}))$

$u = \frac{1}{2} \log(x^2 + y^2)$	$v = \tan^{-1}(\frac{y}{x})$
$u_x = \frac{1}{2} \frac{1}{x^2 + y^2} (2x)$ $= \frac{x}{x^2 + y^2}$ $u_y = \frac{1}{2} \frac{1}{x^2 + y^2} (2y)$ $= \frac{y}{x^2 + y^2}$	$v_x = \frac{1}{1 + \frac{y^2}{x^2}} \left[-\frac{y}{x^2} \right]$ $= \frac{-y}{x^2 + y^2}$ $v_y = \frac{1}{1 + \frac{y^2}{x^2}} \left[\frac{1}{x} \right]$ $= \frac{x}{x^2 + y^2}$

Here, $u_x = v_y$ and $u_y = -v_x$

$\Rightarrow$ C-R equations are satisfied

$\Rightarrow f(z)$ is analytic.

Example: 3.19 Find where each of the following functions ceases to be analytic.

(i) $\frac{z}{(z^2-1)}$ (ii) $\frac{z+i}{(z-i)^2}$

Solution:

(i) Let $f(z) = \frac{z}{(z^2-1)}$

$$f'(z) = \frac{(z^2-1)(1) - z(2z)}{(z^2-1)^2} = \frac{-(z^2+1)}{(z^2-1)^2}$$

$f(z)$ is not analytic, where $f'(z)$ does not exist.

(i.e.) $f'(z) \rightarrow \infty$

(i.e.) $(z^2 - 1)^2 = 0$

(i.e.) $z^2 - 1 = 0$

$$z = 1$$

$$z = \pm 1$$

$\therefore f(z)$ is not analytic at the points $z = \pm 1$

(ii) Let $f(z) = \frac{z+i}{(z-i)^2}$

$$f'(z) = \frac{(z-i)^2(1)(z+i)[2(z-i)]}{(z-i)^4} = \frac{(z+3i)}{(z-i)^3}$$

$$f'(z) \rightarrow \infty, \text{ at } z = i$$

$\therefore f(z)$ is not analytic at $z = i$.

Exercise: 3.1

1. Examine the following function are analytic or not

1. $f(z) = e^x(\cos y + i \sin y)$ [Ans: analytic]
2. $f(z) = e^x(\cos y - i \sin y)$ [Ans: not analytic]
3. $f(z) = z^3 + z$ [Ans: analytic]
4. $f(z) = \sin x \cos y + i \cos x \sinh y$ [Ans: analytic]
5. $f(z) = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$ [Ans: not analytic]
6. $f(z) = 2xy + i(x^2 - y^2)$ [Ans: not analytic]
7. $f(z) = \cosh z$ [Ans: analytic]
8. $f(z) = y$ [Ans: not analytic]
9. $f(z) = (x^2 - y^2 - 2xy) + i(x^2 - y^2 + 2xy)$ [Ans: analytic]
10. $f(z) = \frac{x-iy}{x^2+y^2}$ [Ans: analytic]

2. For what values of z , the function ceases to be analytic.

1. $\frac{1}{z^2-4}$ [Ans: $z = \pm 1$]
2. $\frac{z^2-4}{z^2+1}$ [Ans: $z = \pm 1$]

3. Verify C-R equations for the following functions.

1. $f(z) = ze^z$
2. $f(z) = lz + m$
3. $f(z) = \cos z$

4. Prove that the following functions are nowhere differentiable.

1. $f(z) = e^x(\cos y - i \sin y)$
2. $f(z) = |z|$
3. $f(z) = z - \bar{z}$

5. Find the constants a, b, c so that the following are differentiable at every point.

1. $f(z) = x + ay - i(bx + cy)$ [Ans. $a = b, c = -1$]
2. $f(z) = ax^2 - by^2 + i cxy$ [Ans. $a = b = \frac{c}{2}$]

3.3 PROPERTIES – HARMONIC CONJUGATES

3.3 (a) Laplace equation

$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$ is known as Laplace equation in two dimensions.

3.3 (b) Laplacian Operator

$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is called the Laplacian operator and is denoted by ∇^2 .

Note: (i) $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$ is known as Laplace equation in three dimensions.

Note: (ii) The Laplace equation in polar coordinates is defined as

$$\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

Properties of Analytic Functions

Property: 1 Prove that the real and imaginary parts of an analytic function are harmonic functions.

Proof:

Let $f(z) = u + iv$ be an analytic function

$$u_x = v_y \dots (1) \quad \text{and} \quad u_y = -v_x \quad \dots (2) \text{ by C-R}$$

Differentiate (1) & (2) p.w.r. to x , we get

$$u_{xx} = v_{xy} \dots (3) \quad \text{and} \quad u_{xy} = -v_{xx} \quad \dots (4)$$

Differentiate (1) & (2) p.w.r. to y , we get

$$u_{yx} = v_{yy} \dots (5) \quad \text{and} \quad u_{yy} = -v_{yx} \quad \dots (6)$$

$$(3) + (6) \Rightarrow u_{xx} + u_{yy} = 0 \quad [\because v_{xy} = v_{yx}]$$

$$(5) - (4) \Rightarrow v_{xx} + v_{yy} = 0 \quad [\because u_{xy} = u_{yx}]$$

$\therefore u$ and v satisfy the Laplace equation.

3.3 (c) Harmonic function (or) [Potential function]

A real function of two real variables x and y that possesses continuous second order partial derivatives and that satisfies Laplace equation is called a harmonic function.

Note: A harmonic function is also known as a potential function.

3.3 (d) Conjugate harmonic function

If u and v are harmonic functions such that $u + iv$ is analytic, then each is called the conjugate harmonic function of the other.

Property: 2 If $w = u(x, y) + iv(x, y)$ is an analytic function the curves of the family $u(x, y) = c_1$ and the curves of the family $v(x, y) = c_2$ cut orthogonally, where c_1 and c_2 are varying constants.

Proof:

Let $f(z) = u + iv$ be an analytic function

$$\Rightarrow u_x = v_y \dots (1) \quad \text{and} \quad u_y = -v_x \quad \dots (2) \text{ by C-R}$$

Given $u = c_1$ and $v = c_2$

Differentiate p.w.r. to x , we get

$$u_x + u_y \frac{dy}{dx} = 0 \quad \text{and} \quad v_x + v_y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-u_x}{u_y} \quad \text{and} \quad \frac{dy}{dx} = \frac{-v_x}{v_y}$$

$$\Rightarrow m_1 = \frac{-u_x}{u_y} \quad \Rightarrow m_2 = \frac{-v_x}{v_y}$$

$$m_1 \cdot m_2 = \left(\frac{-u_x}{u_y} \right) \left(\frac{-v_x}{v_y} \right) = \left(\frac{u_x}{u_y} \right) \left(\frac{u_y}{u_x} \right) = -1 \text{ by (1) and (2)}$$

Hence, the family of curves form an orthogonal system.

Property: 3 An analytic function with constant modulus is constant.

Proof:

Let $f(z) = u + iv$ be an analytic function.

$$\Rightarrow u_x = v_y \dots (1) \quad \text{and} \quad u_y = -v_x \dots (2) \text{ by C-R}$$

$$\text{Given } |f(z)| = \sqrt{u^2 + v^2} = c \neq 0$$

$$\Rightarrow |f(z)|^2 = u^2 + v^2 = c^2 \text{ (say)}$$

$$(i.e) u^2 + v^2 = c^2 \dots (3)$$

Differentiate (3) p.w.r. to x and y ; we get

$$2uu_x + 2vv_x = 0 \Rightarrow uu_x + vv_x = 0 \dots (4)$$

$$2uu_y + 2vv_y = 0 \Rightarrow uu_y + vv_y = 0 \dots (5)$$

$$(4) \times u \Rightarrow u^2u_x + uvv_x = 0 \dots (6)$$

$$(5) \times v \Rightarrow uvu_y + v^2v_y = 0 \dots (7)$$

$$(6)+(7) \Rightarrow u^2u_x + v^2v_y + uv[v_x + u_y] = 0$$

$$\Rightarrow u^2u_x + v^2u_x + uv[-u_y + u_y] = 0 \text{ by (1) \& (2)}$$

$$\Rightarrow (u^2 + v^2)u_x = 0$$

$$\Rightarrow u_x = 0$$

Similarly, we get $v_x = 0$

$$\text{We know that } f'(z) = u_x + v_x = 0 + i0 = 0$$

$$\text{Integrating w.r.to } z, \text{ we get, } f(z) = c \quad [\text{Constant}]$$

Property: 4 An analytic function whose real part is constant must itself be a constant. [A.U M/J 2016]

Proof :

Let $f(z) = u + iv$ be an analytic function.

$$\Rightarrow u_x = v_y \dots (1) \quad \text{and} \quad u_y = -v_x \dots (2) \text{ by C-R}$$

$$\text{Given } u = c \quad [\text{Constant}]$$

$$\Rightarrow u_x = 0, \quad u_y = 0$$

$$\Rightarrow u_x = 0, \quad v_x = 0 \quad \text{by (2)}$$

$$\text{We know that } f'(z) = u_x + iv_x = 0 + i0 = 0$$

$$\text{Integrating w.r.to } z, \text{ we get } f(z) = c \quad [\text{Constant}]$$

Property: 5 Prove that an analytic function with constant imaginary part is constant.

Proof:

Let $f(z) = u + iv$ be an analytic function.

$$\Rightarrow u_x = v_y \dots (1) \quad \text{and} \quad u_y = -v_x \dots (2) \text{ by C-R}$$

Given $v = c$ [Constant]

$$\Rightarrow v_x = 0, \quad v_y = 0$$

We know that $f'(z) = u_x + iv_x$

$$= v_y + iv_x \text{ by (1) } = 0 + i0$$

$$\Rightarrow f'(z) = 0$$

Integrating w.r.to z , we get $f(z) = c$ [Constant]

Property: 6 If $f(z)$ and $\overline{f(z)}$ are analytic in a region D , then show that $f(z)$ is constant in that region D .

Proof:

Let $f(z) = u(x, y) + iv(x, y)$ be an analytic function.

$$\overline{f(z)} = u(x, y) - iv(x, y) = u(x, y) + i[-v(x, y)]$$

Since, $f(z)$ is analytic in D , we get $u_x = v_y$ and $u_y = -v_x$

Since, $\overline{f(z)}$ is analytic in D , we have $u_x = -v_y$ and $u_y = v_x$

Adding, we get $u_x = 0$ and $u_y = 0$ and hence, $v_x = v_y = 0$

$$\therefore f(z) = u_x + iv_x = 0 + i0 = 0$$

$$\therefore f(z) \text{ is constant in } D.$$

Problems based on properties

Theorem: 1 If $f(z) = u + iv$ is a regular function of z in a domain D , then $\nabla^2 |f(z)|^2 = 4|f'(z)|^2$

Solution:

Given $f(z) = u + iv$

$$\Rightarrow |f(z)| = \sqrt{u^2 + v^2}$$

$$\Rightarrow |f(z)|^2 = u^2 + v^2$$

$$\Rightarrow \nabla^2 |f(z)|^2 = \nabla^2 (u^2 + v^2)$$

$$= \nabla^2 (u^2) + \nabla^2 (v^2) \quad \dots (1)$$

$$\nabla^2 (u^2) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u^2 = \frac{\partial^2 (u^2)}{\partial x^2} + \frac{\partial^2 (u^2)}{\partial y^2} \quad \dots (2)$$

$$\frac{\partial^2}{\partial x^2} (u^2) = \frac{\partial}{\partial x} \left[2u \frac{\partial u}{\partial x} \right] = 2 \left[u \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} \right] = 2u \frac{\partial^2 u}{\partial x^2} + 2 \left(\frac{\partial u}{\partial x} \right)^2$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2} (u^2) = 2u \frac{\partial^2 u}{\partial y^2} + 2 \left(\frac{\partial u}{\partial y} \right)^2$$

$$(2) \Rightarrow \nabla^2 (u^2) = 2u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right]$$

$$= 0 + 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] \quad [\because u \text{ is harmonic}]$$

$$\nabla^2 (u^2) = 2u_x^2 + 2u_y^2$$

$$\text{Similarly, } \nabla^2 (v^2) = 2v_x^2 + 2v_y^2$$

$$(1) \Rightarrow \nabla^2 |f(z)|^2 = 2[u_x^2 + u_y^2 + v_x^2 + v_y^2]$$

$$= 2[u_x^2 + (-v_x)^2 + v_x^2 + u_x^2] \quad [\because u_x = v_y; u_y = -v_x]$$

$$= 4[u_x^2 + v_x^2]$$

$$(i.e.) \nabla^2 |f(z)|^2 = 4|f'(z)|^2$$

Note : $f(z) = u + iv$; $f'(z) = u_x + iv_x$;

(or) $f'(z) = v_y + iu_y$; $|f'(z)| = \sqrt{u_x^2 + v_x^2}$; $|f'(z)|^2 = u_x^2 + v_x^2$

Theorem: 2 If $f(z) = u + iv$ is a regular function of z in a domain D , then $\nabla^2 \log |f(z)| = 0$ if $f(z) f'(z) \neq 0$ in D . i.e., $\log |f(z)|$ is harmonic in D .

Solution:

$$\text{Given } f(z) = u + iv$$

$$|f(z)| = \sqrt{u^2 + v^2}$$

$$\log |f(z)| = \frac{1}{2} \log (u^2 + v^2)$$

$$\begin{aligned} \nabla^2 \log |f(z)| &= \frac{1}{2} \nabla^2 \log (u^2 + v^2) = \frac{1}{2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log (u^2 + v^2) \\ &= \frac{1}{2} \frac{\partial^2}{\partial x^2} [\log (u^2 + v^2)] + \frac{1}{2} \frac{\partial^2}{\partial y^2} [\log (u^2 + v^2)] \quad \dots (1) \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \frac{\partial^2}{\partial x^2} [\log (u^2 + v^2)] &= \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[\frac{1}{u^2 + v^2} \left(2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} \right) \right] = \frac{\partial}{\partial x} \left[\frac{uu_x + vv_x}{u^2 + v^2} \right] \\ &= \frac{(u^2 + v^2)[uu_{xx} + u_x u_x + vv_{xx} + v_x v_x] - (uu_x + vv_x)(2uu_x + 2vv_x)}{(u^2 + v^2)^2} \\ &= \frac{(u^2 + v^2)[uu_{xx} + vv_{xx} + u_x^2 + v_x^2] - 2(uu_x + vv_x)^2}{(u^2 + v^2)^2} \end{aligned}$$

$$\text{Similarly, } \frac{1}{2} \frac{\partial^2}{\partial y^2} [\log (u^2 + v^2)] = \frac{(u^2 + v^2)[uu_{yy} + vv_{yy} + u_y^2 + v_y^2] - 2(uu_y + vv_y)^2}{(u^2 + v^2)^2}$$

$$\begin{aligned} (1) \Rightarrow \nabla^2 \log |f(z)| &= \frac{(u^2 + v^2)[u(u_{xx} + u_{yy}) + v(v_{xx} + v_{yy}) + (u_x^2 + u_y^2) + (v_x^2 + v_y^2)] - 2[uu_x + vv_x]^2 - 2[uu_y + vv_y]^2}{(u^2 + v^2)^2} \\ &= \frac{(u^2 + v^2)[u(0) + (u_x^2 + v_x^2) + u_y^2 + v_y^2] - 2[u^2 u_x^2 + v^2 v_x^2 + 2uv u_x v_x + u^2 u_y^2 + v^2 v_y^2 + 2uv u_y v_y]}{(u^2 + v^2)^2} \\ &\quad [\because u_{xx} + u_{yy} = 0, v_{xx} + v_{yy} = 0] \\ &= \frac{(u^2 + v^2)[|f'(z)|^2] - 2[u^2 |f'(z)|^2 + v^2 |f'(z)|^2 + 2uv(0)]}{(u^2 + v^2)^2} \end{aligned}$$

$$\begin{aligned} [\because f'(z) = u + iv, |f'(z)|^2 = u_x^2 + v_x^2 \text{ (or) } f'(z) = v_y - iu_y, |f'(z)|^2 = u_y^2 + v_y^2] \\ \text{(or) } |f'(z)|^2 = u_y^2 + v_y^2 \end{aligned}$$

$$= \frac{2(u^2 + v^2)[|f'(z)|^2] - 2[u^2 |f'(z)|^2 + v^2 |f'(z)|^2 + 2uv(0)]}{(u^2 + v^2)^2}$$

$$[\because u_x = v_y, u_y = -v_x]$$

$$\Rightarrow u_x v_x + u_y v_y = 0$$

$$\Rightarrow u_x^2 + u_y^2 = u_x^2 + v_x^2 = |f'(z)|^2$$

$$\Rightarrow v_x^2 + v_y^2 = u_y^2 + v_y^2 = |f'(z)|^2$$

$$= \frac{2(u^2 + v^2)|f(z)|^2 - 2(u^2 + v^2)|f'(z)|^2}{(u^2 + v^2)^2}$$

$$(i.e.) \nabla^2 \log |f(z)| = 0$$

Theorem: 3 If $f(z) = u + iv$ is a regular function of z in a domain D , then

$$\nabla^2(u^p) = p(p-1)u^{p-2}|f'(z)|^2$$

Solution:

$$\nabla^2(u^p) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^p)$$

$$= \frac{\partial^2}{\partial x^2}(u^p) + \frac{\partial^2}{\partial y^2}(u^p)$$

$$\frac{\partial^2}{\partial x^2}(u^p) = \frac{\partial}{\partial x} \left[pu^{p-1} \frac{\partial u}{\partial x} \right] = pu^{p-1}u_{xx} + p(p-1)u^{p-2}(u_x)^2$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2}(u^p) = pu^{p-1}u_{yy} + p(p-1)u^{p-2}(u_y)^2$$

$$(1) \Rightarrow \nabla^2(u^p) = pu^{p-1}(u_{xx} + u_{yy}) + p(p-1)u^{p-2}[u_x^2 + u_y^2]$$

$$= pu^{p-1}(0) + p(p-1)u^{p-2}|f'(z)|^2$$

$$[\because u_{xx} + u_{yy} = 0, f(z) = u + iv, f'(z) = u_x + iv_x, |f'(z)|^2 = u_x^2 + u_y^2]$$

$$\therefore \nabla^2(u^p) = p(p-1)u^{p-2}|f'(z)|^2$$

Theorem: 4 If $f(z) = u + iv$ is a regular function of z , then $\nabla^2|f(z)|^p = p^2|f(z)|^{p-2}|f'(z)|^2$.

Solution:

$$\text{Let } f(z) = u + iv$$

$$|f(z)| = \sqrt{u^2 + v^2} \quad \dots (a)$$

$$|f(z)|^p = (u^2 + v^2)^{p/2} \quad \dots (b)$$

$$\nabla^2|f(z)|^p = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2 + v^2)^{p/2}$$

$$= \frac{\partial^2}{\partial x^2}(u^2 + v^2)^{p/2} + \frac{\partial^2}{\partial y^2}(u^2 + v^2)^{p/2}$$

$$\frac{\partial^2}{\partial x^2}(u^2 + v^2)^{p/2} = \frac{\partial}{\partial x} \left[\frac{p}{2}(u^2 + v^2)^{\frac{p}{2}-1} \left[2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} \right] \right]$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1} [uu_{xx} + u_x u_x + vv_{xx} + v_x v_x]$$

$$+ p \left(\frac{p}{2} - 1 \right) (u^2 + v^2)^{\frac{p}{2}-2} (uu_x + vv_x)(2uu_x + 2vv_x)$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1} [uu_{xx} + u_x^2 + vv_{xx} + v_x^2]$$

$$+ 2p \left(\frac{p}{2} - 1 \right) (u^2 + v^2)^{\frac{p}{2}-2} (uu_x + vv_x)^2$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2}(u^2 + v^2)^{p/2} = p(u^2 + v^2)^{\frac{p}{2}-1} [uu_{yy} + u_y^2 + vv_{yy} + v_y^2]$$

$$+ 2p \left(\frac{p}{2} - 1 \right) (u^2 + v^2)^{\frac{p}{2}-2} (uu_y + vv_y)^2$$

$$\Rightarrow \nabla^2|f(z)|^p = p(u^2 + v^2)^{\frac{p}{2}-1} [u(u_{xx} + u_{yy}) + v(v_{xx} + v_{yy}) + u_x^2 + u_y^2 + v_x^2 + v_y^2] +$$

$$2p \left(\frac{p}{2} - 1 \right) (u^2 + v^2)^{\frac{p}{2}-2} [u^2 u_x^2 + v^2 v_x^2 + 2uv u_x v_x + u^2 u_y^2 + v^2 v_y^2 + 2uv u_y v_y]$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1} [u(0) + v(0) + 2(u_x^2 + u_y^2)]$$

$$\begin{aligned}
& +2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}\left[u^2(u_x^2+u_y^2)+v^2(v_x^2+v_y^2)+2uv(u_xv_x+u_yv_y)\right] \\
& = 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 + 2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}[u^2|f'(z)|^2+v^2|f'(z)|^2+2uv(0)] \\
& = 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 + 2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}(u^2+v^2)|f'(z)|^2 \\
& = 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 + 2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 \\
& = 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2\left[1+\frac{p}{2}-1\right] \\
& = 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 = p^2(u^2+v^2)^{\frac{p-2}{2}}|f'(z)|^2 \\
& = p^2(\sqrt{u^2+v^2})^{p-2}|f'(z)|^2 \\
& = p^2|f(z)|^{p-2}|f'(z)|^2 \text{ by (a) \& (b)}
\end{aligned}$$

Theorem: 5 If $f(z) = u + iv$ is a regular function of z , in a domain D , then

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2 + \left[\frac{\partial}{\partial y}|f(z)|\right]^2 = |f'(z)|^2$$

Solution:

$$\text{Given } f(z) = u + iv$$

$$|f(z)| = \sqrt{u^2 + v^2}$$

$$\frac{\partial}{\partial x}|f(z)| = \frac{\partial}{\partial x}[\sqrt{u^2 + v^2}]$$

$$= \frac{1}{2\sqrt{u^2+v^2}}[2uu_x + 2vv_x] = \frac{uu_x + vv_x}{\sqrt{u^2+v^2}}$$

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2 = \frac{(uu_x + vv_x)^2}{u^2 + v^2} = \frac{u^2u_x^2 + v^2v_x^2 + 2uvu_xv_x}{u^2 + v^2}$$

$$\text{Similarly, } \left[\frac{\partial}{\partial y}|f(z)|\right]^2 = \frac{u^2u_y^2 + v^2v_y^2 + 2uvu_yv_y}{u^2 + v^2}$$

$$\begin{aligned}
\left[\frac{\partial}{\partial x}|f(z)|\right]^2 + \left[\frac{\partial}{\partial y}|f(z)|\right]^2 &= \frac{u^2[u_x^2+u_y^2]+v^2[v_x^2+v_y^2]+2uv[u_xv_x+u_yv_y]}{u^2+v^2} \\
&= \frac{u^2|f'(z)|^2+v^2|f'(z)|^2+2uv \cdot 0}{u^2+v^2} [\because u_x = v_y; u_y = -v_x] \\
&= \frac{(u^2+v^2)|f'(z)|^2}{u^2+v^2} = |f'(z)|^2 [\because u_xv_x + u_yv_y = 0]
\end{aligned}$$

Theorem: 6 If $f(z) = u + iv$ is a regular function of z , then $\nabla^2|\text{Re } f(z)|^2 = 2|f'(z)|^2$

Solution:

$$\text{Let } f(z) = u + iv$$

$$\text{Re } f(z) = u$$

$$|\text{Re } f'(z)|^2 = u^2$$

$$\nabla^2|\text{Re } f'(z)|^2 = \nabla^2 u^2$$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2)$$

$$= \left(\frac{\partial^2}{\partial x^2}\right)(u^2) + \left(\frac{\partial^2}{\partial y^2}\right)(u^2)$$

$$= 2[u_x^2 + u_y^2]$$

$$= 2 |f'(z)|^2$$

Theorem: 7 If $f(z) = u + iv$ is a regular function of z , then prove that $\nabla^2 |\text{Im } f(z)|^2 = 2|f'(z)|^2$

Proof:

$$\text{Let } f(z) = u + iv$$

$$\text{Im } f(z) = v$$

$$|\text{Im } f(z)|^2 = v^2$$

$$\frac{\partial}{\partial x}(v^2) = 2vv_x$$

$$\frac{\partial^2}{\partial x^2}(v^2) = 2[vv_{xx} + v_x v_x] = 2[vv_{xx} + v_x^2]$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2}(v^2) = 2[vv_{yy} + v_y^2]$$

$$\begin{aligned} \therefore \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |\text{Im } f(z)|^2 &= 2[v(v_{xx} + v_{yy}) + v_x^2 + v_y^2] \\ &= 2[v(0) + u_x^2 + v_x^2] \quad \text{by C-R equation} \\ &= 2|f'(z)|^2 \end{aligned}$$

Theorem: 8 Show that $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$ (or) S T $\nabla^2 = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$

Proof:

Let x & y are functions of z and $\bar{z}$

$$\text{that is } x = \frac{z+\bar{z}}{2}, y = \frac{z-\bar{z}}{2i}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x} \frac{\partial x}{\partial z} + \frac{\partial}{\partial y} \frac{\partial y}{\partial z}$$

$$= \frac{\partial}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial}{\partial y} \left[\frac{1}{2i} \right] = \frac{1}{2} \left[\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right]$$

$$2 \frac{\partial}{\partial z} = \frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \quad \dots (1)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial}{\partial y} \frac{\partial y}{\partial \bar{z}}$$

$$= \frac{\partial}{\partial x} \left(\frac{1}{2} \right) + \frac{\partial}{\partial y} \left[\frac{-1}{2i} \right] = \frac{1}{2} \left[\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right]$$

$$2 \frac{\partial}{\partial \bar{z}} = \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \dots (2)$$

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) [\because (a+b)(a-b) = a^2 - b^2]$$

$$= \left(2 \frac{\partial}{\partial z} \right) \left(2 \frac{\partial}{\partial \bar{z}} \right) \text{ by (1) \& (2)}$$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}}$$

Theorem: 9 If $f(z)$ is analytic, show that $\nabla^2 |f(z)|^2 = 4|f'(z)|^2$

Solution:

$$\text{We know that, } \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$$

$$|f(z)|^2 = f(z) \overline{f(z)}$$

$$\begin{aligned}\nabla^2 |f(z)|^2 &= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} [f(z) \overline{f(z)}] \\ &= 4 \left[\frac{\partial}{\partial z} f(z) \right] \left[\frac{\partial}{\partial \bar{z}} \overline{f(z)} \right]\end{aligned}$$

[$\because f(z)$ is independent of $\bar{z}$ and $\overline{f(z)}$ is independent of z]

$$\begin{aligned}\therefore \nabla^2 |f(z)|^2 &= 4 [f'(z) \left[\frac{\partial}{\partial \bar{z}} \overline{f(z)} \right]] = 4 f'(z) \overline{f'(z)} \\ &= 4 |f'(z)|^2 \quad [\because z\bar{z} = |z|^2]\end{aligned}$$

Example: 3.20 Give an example such that u and v are harmonic but $u + iv$ is not analytic.

Solution:

$$u = x^2 - y^2, \quad v = \frac{-y}{x^2 + y^2}$$

Example: 3.21 Find the value of m if $u = 2x^2 - my^2 + 3x$ is harmonic.

Solution:

$$\text{Given } u = 2x^2 - my^2 + 3x$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad [\because u \text{ is harmonic}] \quad \dots (1)$$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= 4x + 3 \\ \frac{\partial^2 u}{\partial x^2} &= 4 \end{aligned} \right| \begin{aligned} \frac{\partial u}{\partial y} &= -2my \\ \frac{\partial^2 u}{\partial y^2} &= -2m \end{aligned}$$

$$\therefore (1) \Rightarrow (4) + (-2m) = 0$$

$$\Rightarrow m = 2$$

3.4 CONSTRUCTION OF ANALYTIC FUNCTION

There are three methods to find $f(z)$.

Method: 1 Exact differential method.

(i) Suppose the harmonic function $u(x, y)$ is given.

Now, $dv = v_x dx + v_y dy$ is an exact differential

Where, v_x and v_y are known from u by using C-R equations.

$$\therefore v = \int v_x dx + \int v_y dy = - \int u_y dx + \int u_x dy$$

(ii) Suppose the harmonic function $v(x, y)$ is given.

Now, $du = u_x dx + u_y dy$ is an exact differential

Where, u_x and u_y are known from v by using C-R equations.

$$\begin{aligned} u &= \int u_x dx + \int u_y dy \\ &= \int v_y dx + \int -v_x dy \\ &= \int v_y dx - \int v_x dy \end{aligned}$$

Method: 2 Substitution method

$$f(z) = 2u \left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a) \right] - [u(a,0), -iv(a,0)]$$

Here, $u(a,0), -iv(a,0)$ is a constant

$$\text{Thus } f(z) = 2u \left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a) \right] + C$$

By taking $a = 0$, that is, if $f(z)$ is analytic $z = 0 + i0$,

We have the simpler formula for $f(z)$

$$f(z) = 2 \left[u \frac{z}{2}, \frac{-iz}{2} \right] + C$$

Method: 3 [Milne – Thomson method]

(i) To find $f(z)$ when u is given

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= u_x - iv_y \text{ [by C-R condition]}$$

$$\therefore f(z) = \int u_x(z,0)dz - i \int u_y(z,0)dz + C \text{ [by Milne–Thomson rule],}$$

Where, C is a complex constant.

(ii) To find $f(z)$ when v is given

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= v_y + iv_x \text{ [by C-R condition]}$$

$$\therefore f(z) = \int v_y(z,0)dz + i \int v_x(z,0)dz + C \text{ [by Milne–Thomson rule],}$$

Where, C is a complex constant.

Example: 3.22 Construct the analytic function $f(z)$ for which the real part is $e^x \cos y$.

Solution:

$$\text{Given } u = e^x \cos y$$

$$\Rightarrow u_x = e^x \cos y \quad [\because \cos 0 = 1]$$

$$\Rightarrow u_x(z,0) = e^x$$

$$\Rightarrow u_y = e^x \cos y \quad [\because \sin 0 = 0]$$

$$\Rightarrow u_y(z,0) = 0$$

$$\therefore f(z) = \int u_x(z,0)dz - i \int u_y(z,0)dz + C \text{ [by Milne–Thomson rule],}$$

Where, C is a complex constant.

$$\therefore f(z) = \int e^z dz - i \int 0 dz + C$$

$$= e^z + C$$

Example: 3.23 Determine the analytic function $w = u + iv$ if $u = e^{2x}(x \cos 2y - y \sin 2y)$

Solution:

$$\text{Given } u = e^{2x}(x \cos 2y - y \sin 2y)$$

$$u_x = e^{2x}[\cos 2y] + (x \cos 2y - y \sin 2y)[2e^{2x}]$$

$$\begin{aligned}
 u_x(z, 0) &= e^{2z}[1] + [z(1) - 0][2e^{2z}] \\
 &= e^{2z} + 2ze^{2z} \\
 &= (1 + 2z)e^{2z}
 \end{aligned}$$

$$u_y = e^{2x}[-2x \sin 2y - (y2 \cos 2y + \sin 2y)]$$

$$u_y(z, 0) = e^{2z}[-0 - (0 + 0)] = 0$$

$$\therefore f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$\begin{aligned}
 f(z) &= \int (1 + 2z)e^{2z}dz - i \int 0 + dz + C \\
 &= \int (1 + 2z)e^{2z}dz + C \\
 &= (1 + 2z)\frac{e^{2z}}{2} - 2\frac{e^{2z}}{4} + C \quad [\because \int uv dz = uv_1 - u'v_2 + u''v_3 - \dots] \\
 &= \frac{e^{2z}}{2} + ze^{2z} - \frac{e^{2z}}{2} + C \\
 &= ze^{2z} + C
 \end{aligned}$$

Example: 3.24 Determine the analytic function where real part is

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1.$$

Solution:

$$\text{Given } u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = 3x^2 - 3y^2 + 6x$$

$$\Rightarrow u_x(z, 0) = 3z^2 - 0 + 6z$$

$$u_y = 0 - 6xy + 0 - 6y$$

$$\Rightarrow u_y(z, 0) = 0$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$\begin{aligned}
 f(z) &= \int (3z^2 + 6z)dz - i \int 0 + dz + C \\
 &= 3\frac{z^2}{3} + 6\frac{z^2}{2} + C \\
 &= z^3 + 3z^2 + C
 \end{aligned}$$

Example: 3.25 Determine the analytic function whose real part in $\frac{\sin 2x}{\cosh 2y - \cos 2x}$

Solution:

$$\text{Given } u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$u_x = \frac{(\cosh 2y - \cos 2x)[2 \cos 2x] - \sin 2x[2 \sin 2x]}{[\cosh 2y - \cos 2x]^2}$$

$$\begin{aligned}
 u_x(z, 0) &= \frac{(1 - \cos 2z)(2 \cos 2z) - 2 \sin^2 2z}{[\cosh 0 - \cos 2z]^2} \\
 &= \frac{2 \cos 2z - 2 \cos^2 2z - 2 \sin^2 2z}{(1 - \cos 2z)^2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \cos 2z - 2[\cos^2 2z + \sin^2 2z]}{(1 - \cos 2z)^2} \\
 &= \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2} \\
 &= \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2} \\
 &= \frac{2 \cos 2z - 2}{(1 - \cos 2z)} \\
 &= \frac{-2}{2 \sin^2 z} \\
 &= -\operatorname{cosec}^2 z
 \end{aligned}$$

$$u_y = \frac{(\cosh 2y - \cos 2x)(0) - \sin 2x[2 \sin 2y]}{[\cosh 2y - \cos 2x]^2}$$

$$\Rightarrow u_y(z, 0) = 0$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}],$$

where C is a complex constant.

$$\begin{aligned}
 f(z) &= \int (-\operatorname{cosec}^2 z) dz - i \int 0 dz + C \\
 &= \cot z + C
 \end{aligned}$$

Example: 3.26 Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic and determine its conjugate.

Also find $f(z)$

Solution:

$$\text{Given } u = \frac{1}{2} \log(x^2 + y^2)$$

$$u_x = \frac{1}{2} \frac{1}{(x^2 + y^2)} (2x) = \frac{x}{x^2 + y^2},$$

$$\Rightarrow u_x(z, 0) = \frac{z}{z^2} = \frac{1}{z}$$

$$u_{xx} = \frac{(x^2 + y^2)[1] - x[2x]}{[x^2 + y^2]^2} = \frac{x^2 + y^2 - 2x^2}{[x^2 + y^2]^2} = \frac{y^2 - x^2}{[x^2 + y^2]^2} \quad \dots (1)$$

$$u_y = \frac{1}{2} \frac{1}{x^2 + y^2} (2y) = \frac{y}{x^2 + y^2}$$

$$\Rightarrow u_y(z, 0) = 0$$

$$u_{yy} = \frac{(x^2 + y^2)[1] - y[2y]}{[x^2 + y^2]^2} = \frac{x^2 - y^2}{[x^2 + y^2]^2} \quad \dots (2)$$

To prove u is harmonic:

$$\therefore u_{xx} + u_{yy} = \frac{(y^2 - x^2) + (x^2 - y^2)}{[x^2 + y^2]^2} = 0 \quad \text{by (1) \& (2)}$$

$\Rightarrow u$ is harmonic.

To find $f(z)$:

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$\begin{aligned}
 f(z) &= \int \frac{1}{z} dz - i \int 0 dz + C \\
 &= \log z + C
 \end{aligned}$$

To find v :

$$f(z) = \log(re^{i\theta}) \quad [\because z = re^{i\theta}]$$

$$u + iv = \log r + \log e^{i\theta} = \log r + i\theta$$

$$\Rightarrow u = \log r, v = \theta$$

Note: $z = x + iy$

$$r = |z| = \sqrt{x^2 + y^2}$$

$$\log r = \frac{1}{2} \log(x^2 + y^2)$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \text{i.e., } v = \tan^{-1}\left(\frac{y}{x}\right)$$

Example: 3.27 Construct an analytic function $f(z) = u + iv$, given that

$u = e^{x^2-y^2} \cos 2xy$. Hence find v .

Solution:

$$\text{Given } u = e^{x^2-y^2} \cos 2xy = e^{x^2} e^{-y^2} \cos 2xy$$

$$u_x = e^{-y^2} [e^{x^2} (-2y \sin 2xy) + \cos 2xy e^{x^2} 2x]$$

$$u_x(z, 0) = 1 [e^{z^2} (0) + 2ze^{z^2}] = 2ze^{z^2}$$

$$u_y = e^{x^2} [e^{-y^2} (-2x \sin 2xy) + \cos 2xy e^{-y^2} (-2y)]$$

$$u_y(z, 0) = e^{z^2} [0 + 0] = 0$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

$$= \int 2ze^{z^2} dz + C$$

$$= 2 \int ze^{z^2} dz + C$$

$$\text{put } t = z^2, dt = 2z dz$$

$$= \int e^t dt + C$$

$$= e^t + C$$

$$f(z) = e^{z^2} + C$$

To find v :

$$u + iv = e^{(x+iy)^2} = e^{x^2-y^2+i2xy} = e^{x^2-y^2} e^{i2xy}$$

$$= e^{x^2-y^2} [\cos(2xy) + i \sin(2xy)]$$

$$v = e^{x^2-y^2} \sin 2xy \quad [\because \text{equating the imaginary parts}]$$

Example: 3.28 Find the regular function whose imaginary part is

$e^{-x}(x \cos y + y \sin y)$.

Solution:

$$\text{Given } v = e^{-x}(x \cos y + y \sin y)$$

$$v_x = e^{-x} [\cos y] + (x \cos y + y \sin y) [-e^{-x}]$$

$$v_x(z, 0) = e^{-z} + (z)(-e^{-z}) = (1 - z)e^{-z}$$

$$v_y = e^{-x}[-x \sin y + (y \cos y + \sin y (1))]$$

$$v_x(z, 0) = e^{-z}[0 + 0 + 0] = 0$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$\begin{aligned} f(z) &= \int 0dz + i \int (1 - z)e^{-z}dz + C \\ &= i \int (1 - z)e^{-z}dz + C \\ &= i \left[(1 - z) \left[\frac{e^{-z}}{-1} \right] - (-1) \left[\frac{e^{-z}}{(-1)^2} \right] \right] + C \\ &= i[-(1 - z)e^{-z} + e^{-z}] + C \\ &= ize^{-z} + C \end{aligned}$$

Example: 3.29 In a two dimensional flow, the stream function is $\psi = \tan^{-1}\left(\frac{y}{x}\right)$. Find the velocity potential ϕ .

Solution:

$$\text{Given } \psi = \tan^{-1}(y/x)$$

We should denote, ϕ by u and ψ by v

$$\therefore v = \tan^{-1}(y/x)$$

$$v_x = \frac{1}{1+(y/x)^2} \left[\frac{-y}{x^2} \right] = \frac{-y}{x^2+y^2},$$

$$v_x(z, 0) = 0$$

$$v_y = \frac{1}{1+(y/x)^2} \left[\frac{1}{x} \right] = \frac{x}{x^2+y^2}$$

$$v_x(z, 0) = \frac{z}{z^2} = \frac{1}{z}$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C$$

$$f(z) = \int \frac{1}{z} dz + i \int 0 dz + C = \log z + C$$

To find ϕ :

$$f(z) = \log(re^{i\theta}) \quad [\because z = re^{i\theta}]$$

$$u + iv = \log r + \log e^{i\theta}$$

$$u + iv = \log r + i\theta$$

$$\Rightarrow u = \log r$$

$$\Rightarrow u = \log \sqrt{x^2 + y^2}$$

$$= \frac{1}{2} \log(x^2 + y^2)$$

$$z = x + iy, |z| = \sqrt{x^2 + y^2}$$

So, the velocity potential ϕ is

$$\phi = \frac{1}{2} \log(x^2 + y^2)$$

Note: In two dimensional steady state flows, the complex potential

$f(z) = \phi(x, y) + i\psi(x, y)$ is analytic.

Example: 3.30 If $w = u + iv$ is an analytic function and $v = x^2 - y^2 + \frac{x}{x^2+y^2}$, find u .

Solution:

$$\text{Given } v = x^2 - y^2 + \frac{x}{x^2+y^2}$$

$$v_x = 2x - 0 + \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2}$$

$$= 2x + \frac{y^2 - x^2}{(x^2+y^2)^2}, \quad v_x(z, 0) = 2z + \frac{(-z^2)}{(z^2)}$$

$$\Rightarrow v_x(z, 0) = 2z - \frac{1}{z^2}$$

$$v_y = 0 - 2y + \frac{0 - x(2y)}{(x^2+y^2)^2}$$

$$= 0 - 2y - \frac{2xy}{(x^2+y^2)^2}$$

$$\Rightarrow v_y(z, 0) = 0$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int 0dz + i \int \left(2z - \frac{1}{z^2}\right) dz + C$$

$$= i \left[2 \frac{z^2}{2} + \frac{1}{z} \right] + C \quad \left[\because \int \frac{-1}{z^2} dz = \frac{1}{z} \right]$$

$$= i \left[z^2 + \frac{1}{z} \right] + C$$

Example: 3.31 If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .

Solution:

$$\text{Given } u - v = e^x(\cos y - \sin y), \quad \dots (A)$$

Differentiate (A) p.w.r. to x , we get

$$u_x - v_x = e^x(\cos y - \sin y),$$

$$u_x(z, 0) - v_x(z, 0) = e^z \quad \dots (1)$$

Differentiate (A) p.w.r. to y , we get

$$u_y - v_y = e^x(-\sin y - \cos y)$$

$$u_y(z, 0) - v_y(z, 0) = e^z[-1]$$

$$\text{i.e., } u_y(z, 0) - v_y(z, 0) = -e^z$$

$$-v_x(z, 0) - u_x(z, 0) = -e^z \quad \dots (2) \quad [\text{by C-R conditions}]$$

$$(1) + (2) \Rightarrow -2v_x(z, 0) = 0$$

$$\Rightarrow v_x(z, 0) = 0$$

$$(1) \Rightarrow u_x(z, 0) = e^z$$

$$f(z) = \int u_x(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

$$f(z) = \int e^z dz + i0 + C$$

$$= e^z + C$$

Example: 3.32 Find the analytic functions $f(z) = u + iv$ given that

(i) $2u + v = e^x(\cos y - \sin y)$

(ii) $u - 2v = e^x(\cos y - \sin y)$

Solution:

Given (i) $2u + v = e^x(\cos y - \sin y) \quad \dots (A)$

Differentiate (A) p.w.r. to x, we get

$$2u_x + v_x = e^x(\cos y - \sin y)$$

$$2u_x - u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$2u_x(z, 0) - u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate (A) p.w.r. to y, we get

$$2u_y + v_y = e^x[-\sin y - \cos y]$$

$$2u_y + u_x = e^x[-\sin y - \cos y] \quad [\text{by C-R condition}]$$

$$2u_y(z, 0) + u_x(z, 0) = e^z(-1) = -e^z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 4u_x(z, 0) - 2u_y(z, 0) = 2e^z \quad \dots (3)$$

$$(2) + (3) \Rightarrow 5u_x(z, 0) = e^z$$

$$\Rightarrow u_x(z, 0) = \frac{1}{5}e^z$$

$$(1) \Rightarrow u_y(z, 0) = \frac{2}{5}e^z - e^z = -\frac{3}{5}e^z$$

$$\Rightarrow u_y(z, 0) = -\frac{3}{5}e^z$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int \frac{1}{5}e^z dz - i \int -\frac{3}{5}e^z dz + C$$

$$= \frac{2}{5}e^z + \frac{3}{5}ie^z + C$$

$$= \frac{1+3i}{5}e^z + C$$

(ii) $u - 2v = e^x(\cos y - \sin y) \quad \dots (B)$

Differentiate (B) p.w.r. to x, we get

$$u_x - 2v_x = e^x(\cos y - \sin y)$$

$$u_x + 2u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$u_x(z, 0) + 2u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate (B) p.w.r. to y, we get

$$u_y - 2v_y = e^x[-\sin y - \cos y]$$

$$u_y - 2u_x = e^x[-\sin y - \cos y] \quad [\text{by C-R condition}]$$

$$u_y(z, 0) - 2u_x(z, 0) = -e^z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 2u_x(z, 0) + 4u_y(z, 0) = 2e^z \quad \dots (3)$$

$$(2) + (3) \Rightarrow 5u_y(z, 0) = e^z$$

$$\Rightarrow u_y(z, 0) = \frac{1}{5} e^z$$

$$(1) \Rightarrow u_x(z, 0) = -\frac{2}{5} e^z + e^z \\ = \frac{3}{5} e^z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int \frac{3}{5} e^z dz - i \int \frac{1}{5} e^z dz + C \\ = \frac{3}{5} e^z - i \frac{1}{5} e^z + C = \frac{3-i}{5} e^z + C$$

Example: 3.33 Determine the analytic function $f(z) = u + iv$ given that

$$3u + 2v = y^2 - x^2 + 16xy$$

Solution:

$$\text{Given } 3u + 2v = y^2 - x^2 + 16xy \quad \dots (A)$$

Differentiate (A) p.w.r. to x, we get

$$3u_x + 2v_x = -2x + 16y$$

$$3u_x - 2u_y = -2x + 16y \quad [\text{by C-R condition}]$$

$$3u_x(z, 0) - 2u_y(z, 0) = -2z \quad \dots (1)$$

Differentiate (A) p.w.r. to y, we get

$$3u_y + 2v_y = 2y + 16x$$

$$3u_y + 2u_x = 2y + 16x \quad [\text{by C-R condition}]$$

$$3u_y(z, 0) + 2u_x(z, 0) = 16z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 6u_x(z, 0) - 4u_y(z, 0) = -4z \quad \dots (3)$$

$$(2) \times (3) \Rightarrow 9u_y(z, 0) + 6u_x(z, 0) = 48z$$

$$(3) - (4) \Rightarrow -13u_y(z, 0) = -52z$$

$$\Rightarrow u_y(z, 0) = 4z$$

$$(1) \Rightarrow 3u_x(z, 0) = 8z - 2z = 6z$$

$$\Rightarrow u_x(z, 0) = 2z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

where C is a complex constant.

$$f(z) = \int 2z dz - i \int 4z dz + C \\ = 2 \frac{z^2}{2} - i \frac{4z^2}{2} + C \\ = z^2 - i2z^2 + C \\ = (1 - i2)z^2 + C$$

Example:3.34 Find an analytic function $f(z) = u + iv$ given that $2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$

Solution:

$$\text{Given } 2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

Differentiate p.w.r. to x , we get

$$2u_x + 3v_x = \frac{(\cosh 2y - \cos 2x)(2 \cos 2x) - \sin 2x (2 \sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$$2u_x - 3u_y = \frac{(\cosh 2y - \cos 2x)(2 \cos 2x) - \sin 2x (2 \sin 2x)}{(\cosh 2y - \cos 2x)^2} \quad [\text{by C-R condition}]$$

$$\begin{aligned} 2u_x(z, 0) - 3u_y(z, 0) &= \frac{2 \cos 2z(1 - \cos 2z) - 2 \sin^2 2z}{(1 - \cos 2z)^2} \\ &= \frac{2 \cos 2z - 2 \cos^2 2z - 2 \sin^2 2z}{(1 - \cos 2z)^2} \end{aligned}$$

$$= \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z}$$

$$= \frac{-2}{2 \sin^2 z} = -\operatorname{cosec}^2 z$$

$$2u_x(z, 0) - 3u_y(z, 0) = -\operatorname{cosec}^2 z \quad \dots (1)$$

Differentiate p.w.r. to y , we get

$$2u_y + 3v_y = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2} \quad (2)$$

$$2u_y + 3u_x = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2} \quad [\text{by C - R condition}]$$

$$2u_y(z, 0) + 3u_x(z, 0) = 0 \quad \dots (2)$$

Solving (1) & (2) we get,

$$\Rightarrow u_x(z, 0) = -\frac{2}{13} \operatorname{cosec}^2 z$$

$$\Rightarrow u_y(z, 0) = -\frac{2}{13} \operatorname{cosec}^2 z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant

$$\begin{aligned} f(z) &= \int \left(\frac{-2}{13} \right) \operatorname{cosec}^2 z \, dz - i \int \left(\frac{3}{13} \right) \operatorname{cosec}^2 z \, dz + C \\ &= \left(\frac{2}{13} \right) \cot z + \left(\frac{3}{13} \right) \cot z + C \\ &= \frac{2+3i}{13} \cot z + C \end{aligned}$$

Example: 3.35 Find the analytic function $f(z) = u + iv$ given that $2u + 3v = e^x(\cos y - \sin y)$

Solution:

$$\text{Given } 2u + 3v = e^x(\cos y - \sin y)$$

Differentiate p.w.r. to x , we get

$$2u_x + 3v_x = e^x(\cos y - \sin y)$$

$$2u_x - 3u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$2u_x(z, 0) - 3u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate p.w.r. to y, we get

$$2u_y + 3v_y = e^x[-\sin y - \cos y]$$

$$2u_y + 3u_x = -e^x [\sin y + \cos y] \quad [\text{by C-R condition}]$$

$$2u_y(z, 0) + 3u_x(z, 0) = -e^z \quad \dots (2)$$

$$(1) \times (3) \Rightarrow 6u_x(z, 0) - 9u_y(z, 0) = 3e^z \quad \dots (3)$$

$$(2) \times 2 \Rightarrow 6u_x(z, 0) + 4u_y(z, 0) = -2e^z \quad \dots (4)$$

$$(3) - (4) \Rightarrow -13u_y(z, 0) = 5e^z$$

$$\Rightarrow u_y(z, 0) = -\frac{5}{13}e^z$$

$$(1) \Rightarrow 2u_x(z, 0) + \frac{15}{13}e^z = e^z$$

$$2u_x(z, 0) = e^z - \frac{15}{13}e^z = -\frac{2}{13}e^z$$

$$\Rightarrow u_x(z, 0) = -\frac{1}{13}e^z$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C$$

$$\therefore f(z) = \int \frac{-1}{13}e^z dz - i \int \left(\frac{-5}{13}\right)dz + C$$

$$= \frac{-1}{13}e^z + \frac{5}{13}e^z i + C = \frac{-1+5i}{13}e^z + C$$

Exercise: 3.4

Construction of an analytic function

1. Show that the function $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Find the conjugate harmonic function v and express $u + iv$ as an analytic function of z .

[Ans: $v(x, y) = 3x^2y + 4xy - x^3 + C$, $f(z) = -iz^3 + 2z^2 + iC$ where C is a real constant.]

2. If $f(z) = u + iv$ is an analytic function of z , and if $u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$, find v .

$$[\text{Ans: } v = \frac{-2 \sinh 2y}{e^{2y} + e^{-2y} - 2 \cos 2x} + C]$$

3. Find v such that $w = u + iv$ is an analytic function of z , given that $u = e^{x^2-y^2} \cos 2xy$. Hence find w .

$$[\text{Ans: } v = e^{x^2-y^2} \sin 2xy + C \quad w = e^{z^2} + C]$$

4. Find the analytic function $w = u + iv$ if $w = e^{2x}(x \cos 2y - y \sin 2y)$. Hence find u .

$$[\text{Ans: } w = iz e^{2z} + C, u = -(x \sin 2y + y \cos 2y) e^{2x} + C]$$

5. If $v = \frac{x-y}{x^2+y^2}$ find u such $u + iv$ is an analytic function. What is the harmonic conjugate of v ?

$$[\text{Ans: } u = \frac{x+y}{x^2+y^2} + C \quad \text{Harmonic conjugate of } v \text{ is } -u = \frac{-(x+y)}{x^2+y^2}, f(z) = \frac{1+i}{z} + C]$$

6. Find the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y + \cos 2x}$

$$[\text{Ans: } f(z) = \tan z + C]$$

7. Find the analytic function whose imaginary part is $-e^{-2xy} \cos(x^2 - y^2)$

$$[\text{Ans: } f(z) = -ie^{iz^2} + C]$$

6. Prove that $u = 2^x - x^3 + 3xy^2$ is harmonic and find its harmonic conjugate. Also find the corresponding analytic function. [Ans: $v = 2y - 3x^2y + y^3 + C, f(z) = 2z - z^3 + iC$]

7. Find the real part of the analytic function whose imaginary part is $e^{-x}[2xy \cos y + (y^2 - x^2) \sin y]$.

Construct the analytic function.

$$[\text{Ans: } u = e^{-x}[(x^2 - y^2) \cos y + 2xy \sin y], f(z) = z^2 e^{-z} + C]$$

8. Find the analytic function $f(z) = u + iv$ given that $2u + v = e^{2x}[(2x + y) \cos 2y + (x - 2y) \sin 2y]$

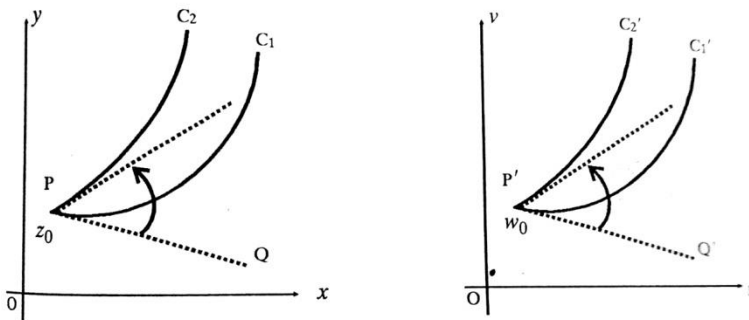
$$[\text{Ans: } f(z) = ze^{2z} + C]$$

9. Prove that $u = x^2 - y^2$ and $v = -\frac{y}{x^2 + y^2}$ are harmonic functions but not harmonic conjugates.

3.5 CONFORMAL MAPPING

Definition: Conformal Mapping

A transformation that preserves angles between every pair of curves through a point, both in magnitude and sense, is said to be conformal at that point.



Definition: Isogonal

A transformation under which angles between every pair of curves through a point are preserved in magnitude, but altered in sense is said to be an isogonal at that point.

Note: 3.4 (i) A mapping $w = f(z)$ is said to be conformal at $z = z_0$, if $f'(z_0) \neq 0$.

Note: 3.4 (ii) The point, at which the mapping $w = f(z)$ is not conformal,

(i.e.) $f'(z) = 0$ is called a **critical point** of the mapping.

If the transformation $w = f(z)$ is conformal at a point, the inverse transformation $z = f^{-1}(w)$ is also conformal at the corresponding point.

The critical points of $z = f^{-1}(w)$ are given by $\frac{dz}{dw} = 0$. hence the critical point of the transformation $w = f(z)$ are given by $\frac{dw}{dz} = 0$ and $\frac{dz}{dw} = 0$,

Note: 3.4 (iii) Fixed points of mapping.

Fixed or invariant point of a mapping $w = f(z)$ are points that are mapped onto themselves, are “Kept fixed” under the mapping. Thus they are obtained from $w = f(z) = z$.